

Project Assignment: Fullstack Web Application

Project Overview

In this project, you and your team will design and implement a fullstack web application using **Spring Boot** for the backend and **ionic with Vue.js (v3)** for the frontend.

You will apply the principles of **Clean Architecture** (as discussed in class), develop a **mobile-first UI**, and organize your work using **GitHub Classroom** and a **Kanban board** for backlog and task management.

The goal is not only to develop working software but to show thoughtful **design**, **good engineering practices**, and **clean implementation**.

Team Setup

- Form teams of **4 students**
- One GitHub Classroom repository per team (created via invitation link)
- Use GitHub Projects (or similar) to maintain a **Kanban board**
- Every team member must contribute meaningfully and visibly to the code

Project Scope

Each team will:

1. Design the **database schema**, **REST API** (resource-based, Richardson Model and a **paper prototype** for the mobile UI
2. Implement the application in two stages (milestones)
3. Perform **peer reviews** to improve quality and get external feedback
4. document AI usage and workflows

Technology Stack

- **Backend:** Java with Spring Boot
 - Follow Clean Architecture (as shown in the lectures etc.)
- **Frontend:** Ionic with Vue 3
 - Use modular design (e.g. Atomic Design or similar)
 - Maintain state with Pinia
 - Use Axios Library or JS fetch API for Ajax Calls
- **Tooling:**
 - GitHub Classroom
 - GitHub Projects (Kanban)
 - Figma (optional) for digital prototypes
 - Markdown for documentation

Project Milestones

Milestone 1 – *Planning & Specification*

Deliverables:

- REST API specification (resource-oriented, Richardson Maturity Model)
- Database model (ER diagram or similar)
- Backlog (Kanban board with user stories as cards) or similar project management
- Peer review of another group's Milestone 1

Milestone 2 – *Partial Implementation*

Deliverables:

- Fullstack implementation of **user stories 1–3**
 - Must include working backend and frontend
 - Frontend should follow **modular design** and **clean architecture**
 - Use state management where applicable (e.g. for selected items, modals, forms)
- Updated REST API docs and DB model if needed
- Updated Kanban board (reflect progress)
- Peer review of another group's Milestone 2

Final Delivery – *Completion & Quality*

Deliverables:

- Full implementation of **user stories 4–5**
- UI/UX quality and usability
- Clean code, good practices, proper error handling, validations
- Short README with project summary and how to run it
- Presentation of your App in class

Submission

All deliverables must be submitted via your **GitHub Classroom repository**. Make sure to use **issues and pull requests** to organize your collaboration.

The **Kanban board** must reflect real progress, planning, and review steps.

AI Usage Policy

You are encouraged to use AI tools as part of your development workflow.

However, the following rules apply:

- Every meaningful use of AI must be documented in a dedicated `AI_USAGE.md` file in your repository
- Document: which tool, what prompt or task, what was accepted/modified/rejected, and why
- You must be able to explain and defend every part of your code in a peer review in class and final presentation
- Undocumented AI use that becomes apparent during review counts as academic dishonesty

Pro Tips

- Think about how to arrange the work in your team first
- Start small and iterate! Avoid overly complex ideas.
- Keep your code clean and modular – follow the patterns from class.
- Commit often and meaningfully.
- Help your peers – your reviews matter.
- Build something you'd enjoy using.

Happy Coding!